DO NOW, AO1 REVISION

Mains electricity and the National Grid

Electricity
Paper 1, Combined and Separate
32 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 15 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**What do a.c. and d.c. stand for?**	Alternating current and direct current	☐
2 ★	**What is an alternating current (a.c.)?**	A current that repeatedly reverses direction.	☐
3 ★	**State the frequency and potential difference of the UK mains electricity supply.**	50 Hz and about 230 V	☐
4 ★	**State the colour of the insulation on the live, neutral and earth wires.**	Live: brown Neutral: blue Earth: green and yellow stripes	☐
5 ★	**State the function of the live wire in a three-core cable.**	It carries the alternating potential difference from the supply.	☐
6 ★	**State the function of the neutral wire in a three-core cable.**	It completes the circuit.	☐
7 ★	**State the function of the earth wire in a three-core cable.**	It is a safety wire that stops the appliance becoming live.	☐
8	**State the potential difference between the live wire and the earth wire.**	About 230 V	☐
9 ★	**Name the wires in a three-core cable that are at, or close to, 0 V.**	The earth wire and the neutral wire	☐
10	**State when the earth wire will carry a current.**	Only when there is a fault.	☐
11 ★	**In a plug, name the wire that is connected to the fuse.**	The live wire	☐
12 ★	**Explain why touching the live wire is dangerous even when the appliance is switched off.**	The live wire is still at about 230 V, so a current would flow through the person to earth.	☐

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	QUESTION	ANSWER	✓
13	**Define power in terms of energy.**	The rate at which energy is transferred.	☐
14 ★	**State the standard unit of power.**	Watt (W)	☐
15	**Give two things that affect the energy transferred by an appliance.**	Its power and the time it is switched on	☐
16	**What does a higher power rating tell you about an appliance?**	It transfers more energy each second.	☐
17 ★	**Describe the National Grid and its purpose.**	A system of cables and transformers linking power stations to consumers.	☐
18	**State what is transferred across the National Grid.**	Electrical power	☐
19	**State the function of transformers.**	To increase or decrease the potential difference.	☐
20	**What does a step-up transformer do to the potential difference?**	It increases it.	☐
21 ★	**What does a step-up transformer do to the current?**	It decreases it.	☐
22 ★	**Explain why step-up transformers are used before electricity is transmitted.**	A higher potential difference means a smaller current, so less energy is wasted heating the cables.	☐
23 ★	**What is direct current (d.c.)?**	A current that flows in one direction only.	☐
24	**Name a source of direct current.**	A cell or battery	☐
25	**State the equation linking power, potential difference and current.**	power = potential difference × current ($P = V\ I$)	☐
26	**State the equation linking power, current and resistance.**	power = current^2 × resistance ($P = I^2\ R$)	☐
27	**State the equation linking energy transferred, power and time.**	energy transferred = power × time ($E = P\ t$)	☐
28	**State the equation linking energy transferred, charge flow and potential difference.**	energy transferred = charge flow × potential difference ($E = Q\ V$)	☐

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	QUESTION	ANSWER	✓
29	**State the equation for a transformer that is 100% efficient.**	p.d. × current in the secondary = p.d. × current in the primary ($V_s \times I_s = V_p \times I_p$)	☐
30	**A battery-powered fan is switched on. Describe the useful energy transfer.**	Energy is transferred electrically from the chemical store of the battery to the kinetic store of the fan.	☐
31 ★	**Explain why step-down transformers are used before electricity reaches homes.**	To reduce the potential difference to a safer value (230 V).	☐
32	**Explain why any connection between the live wire and earth is dangerous.**	A very large current would flow, which could cause a fire or a fatal shock.	☐

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